Triangles

The big idea

Two figures are **congruent** if they have the same shape and size. They are **similar** if they have the same **shape** but not necessarily the same size, like a photo and its enlargement. Similarity lets us find heights of trees, widths of rivers and distances we cannot measure directly.

SIMILAR TRIANGLES

△ABC ~ △DEF means: corresponding angles are equal (∠A = ∠D, ∠B = ∠E, ∠C = ∠F) **and** corresponding sides are in the same ratio: AB/DE = BC/EF = CA/FD.

All circles are similar. All squares are similar. All equilateral triangles are similar. But two rectangles or two rhombuses need not be.

Basic Proportionality Theorem (Thales' theorem)

BPT

If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, it divides those two sides in the **same ratio**: if DE ∥ BC in △ABC, then AD/DB = AE/EC.

Its **converse** is also true: if a line divides two sides of a triangle in the same ratio, it is parallel to the third side. The proof of BPT (using areas of triangles with equal heights) is a frequent 5–mark question.

IN △ABC, DE ∥ BC, AD = 1.5 CM, DB = 3 CM AND AE = 1 CM. FIND EC.

By BPT, AD/DB = AE/EC ⇒ 1.5/3 = 1/EC ⇒ **EC = 2 cm**.

Criteria for similarity

Criterion	What must match
AAA (or AA)	Two angles of one triangle equal two angles of the other
SSS	All three sides in the same ratio
SAS	One angle equal and the sides including it in the same ratio

AA IS ENOUGH

If two angles match, the third must match too (angle sum is 180°). So in proofs, finding two equal angles is usually the quickest route.

A 6 M POLE CASTS A 4 M SHADOW; AT THE SAME TIME A TOWER CASTS A 28 M SHADOW. FIND THE TOWER'S HEIGHT.

The sun's rays make equal angles, so the triangles are similar (AA).
h/28 = 6/4 ⇒ **h = 42 m**.

WRITING VERTICES IN THE WRONG ORDER

In △ABC ~ △PQR the order matters: A ↔ P, B ↔ Q, C ↔ R. Pair the sides in that same order.

Quick recap

- Similar = same shape; equal angles and proportional sides.
- BPT: a line parallel to one side divides the other two proportionally (and its converse).
- Tests: AA, SSS, SAS.

Practice questions

1. Which pair is always similar?
   (a) Two rectangles (b) Two rhombuses (c) Two equilateral triangles (d) Two isosceles triangles
2. In $\triangle ABC$, $DE \parallel BC$ with $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm. Then EC is
   (a) 5 cm (b) 6 cm (c) 8 cm (d) 2.67 cm
3. If $\triangle ABC \sim \triangle PQR$ with $AB/PQ = 2/3$, then BC/QR is
   (a) 2/3 (b) 3/2 (c) 4/9 (d) 1
4. Two triangles have two pairs of equal angles. They are
   (a) congruent (b) similar (c) neither (d) right-angled
5. The converse of BPT tells us that if $AD/DB = AE/EC$ in $\triangle ABC$, then
   (a) $AD = AE$ (b) $DE \parallel BC$ (c) $DE = BC$ (d) $\triangle ADE \cong \triangle ABC$

Answers: 1. c 2. b 3. a 4. b 5. b